

JeonStack: Stacking Korean Folktales into a Live Narrative System

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Figure 1: Overview of the JeonStack installation.

Abstract

We present *JeonStack*, an interactive storytelling installation that reinterprets Korean folktales as modular narrative units activated through physical book stacking. Using nine *unmodified* folktale books, the system detects the order of stacked books via visual markers and composes a live audiovisual scene consisting of animated backgrounds, environment-mapped character actions, synthesized dialogue, and reactive *world transitions*. Rather than encouraging reading, JeonStack introduces stacking as a narrative grammar in which selection and order determine how worlds and characters are recombined through embodied interaction.

CCS Concepts

• Human-centered computing → Tangible user interfaces; Interactive systems and tools; • Applied computing → Arts and humanities; • Computing methodologies → Artificial intelligence.

Keywords

Tangible interaction; interactive narrative; Korean folktales; book-based interface; audiovisual storytelling

1 Introduction

Korean folktales—many titled with the suffix *-jeon*—have long been experienced as fixed narratives, read linearly and consumed as complete, self-contained stories. Even when adapted into digital

media, these tales typically preserve a predetermined structure, allowing audiences to choose *which* story to engage with while leaving the internal narrative sequence unchanged. As a result, interaction is often limited to selection at the level of titles rather than composition at the level of narrative structure.

This fixed mode of engagement poses a limitation for interactive storytelling. While prior work has explored narrative as something that can be authored through spatial arrangement and interaction [1], most adaptations of folktales continue to treat stories as indivisible units. Audiences remain positioned as readers or viewers of a completed text, with little opportunity to actively reconfigure how different story worlds, characters, or situations relate to one another.

JeonStack addresses this gap by asking: *What if folktales were not read, but stacked?* Rather than rewriting or digitizing stories, JeonStack preserves existing books as cultural artifacts and redesigns the interaction around them. By introducing physical book stacking as the primary mode of interaction, the system enables users to compose narratives through selection and order, allowing worlds and characters to be recombined through embodied interaction.

2 JeonStack

2.1 Books as modular narrative units

JeonStack uses nine Korean folktale books—*Simcheongjeon*, *Heungbujeon*, *Onggojipjeon*, *Dukkeopjeon*, *Janghwa Hongryeonjeon*, *Kimwonjeon*, *Byeoljubujeon*, *Baksijeon*, and *Jeon Woochi-jeon*—borrowed from a library. All books belong to the same publication series,

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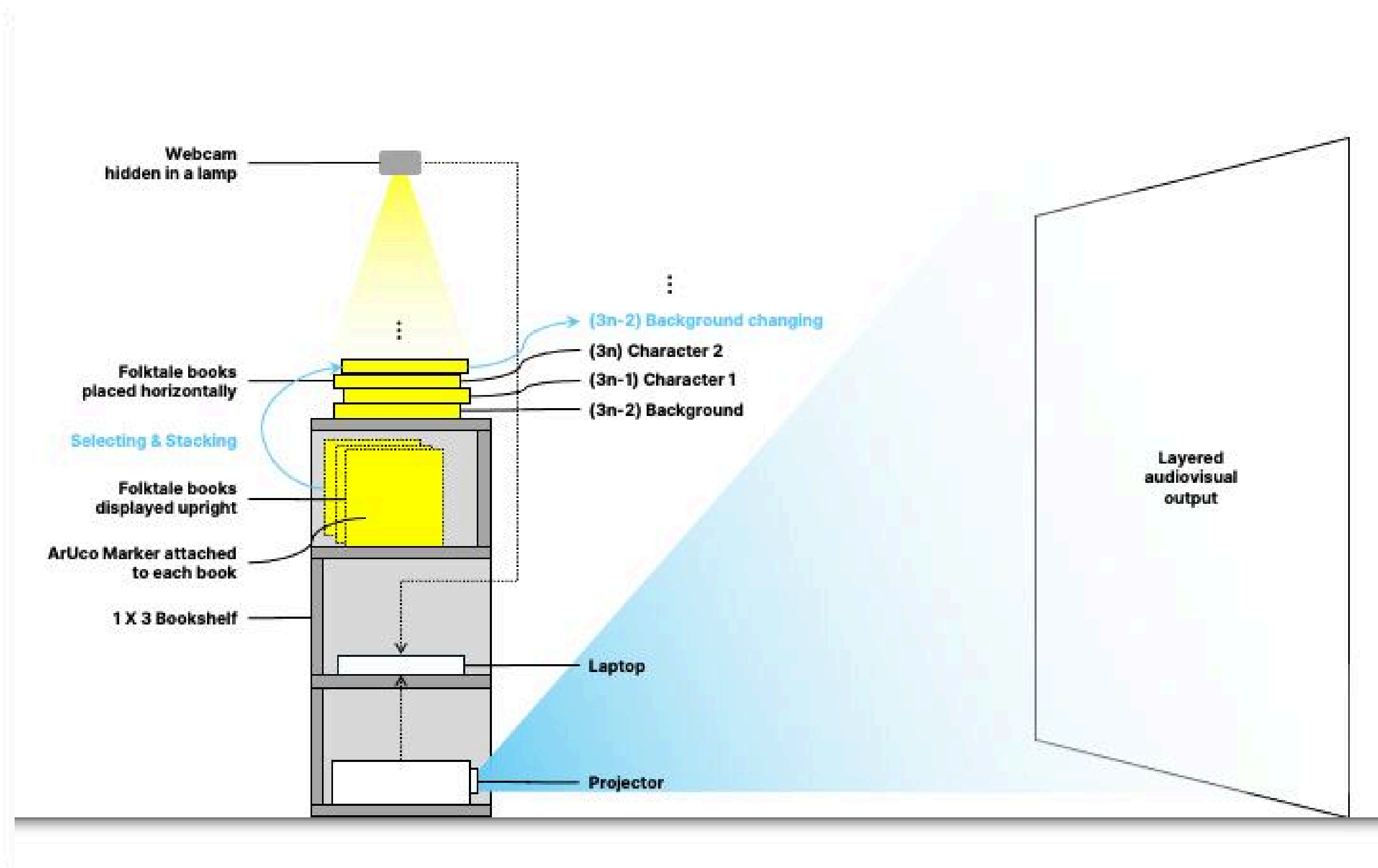


Figure 2: System diagram of JeonStack.

resulting in uniform size and materiality, while featuring illustrations by different authors. Importantly, the books are left entirely unmodified; their texts and images remain intact and readable.

Each book cover carries a small fiducial marker for rapid recognition. We use ArUco-style markers [2] to enable robust detection within approximately one second under typical indoor lighting and viewing angles.

2.2 Interaction: selection → stacking

The installation separates *selection* from *composition*. Books are displayed upright on the top shelf like ordinary library volumes. Interaction begins when a participant selects a book by pulling it out from the shelf and places it horizontally on top of the bookshelf. This act of stacking—not browsing or reading—drives the narrative system. By requiring users to remove books from the shelf before stacking them, JeonStack emphasizes selection as an integral part of narrative composition and positions books as a tangible interface [3].

2.3 Narrative state machine (3n-2 / 3n-1 / 3n)

JeonStack operates as a repeatable narrative state machine with three persistent slots: one background and two character roles (Character 1 and Character 2).

(3n-2) Background: The first stacked book initializes a folktale-specific background world (looping background video) and background music.

(3n-1) Character 1: The second stacked book fills the Character 1 slot. The character performs an *environment-mapped action* (e.g., on the *Simcheongjeon* sea-ship background, diving into the sea) while delivering a spoken line conditioned on both the character profile and the current world.

(3n) Character 2: The third stacked book fills the Character 2 slot, producing a short dialogue exchange shaped by the shared background and the counterpart’s persona.

When additional books are stacked, the system updates characters through *replacement rather than accumulation*: each new character input replaces the oldest character occupying the same role, and interactions are regenerated for the current configuration. When a new background book is stacked, a *world transition*

occurs. The background changes abruptly while existing characters remain on screen and generate reactive behaviors and utterances—such as surprise or confusion—as if they have been instantaneously transported into a different environment.

If users remove books mid-sequence, the system updates based on the *topmost remaining* detected marker; at present, these changes are treated as immediate state updates rather than intentional narrative events. If no marker is detected for five seconds, the system fades to black and resets.

2.4 Character profiles and dialogue generation

Each character is defined through a compact profile—including gender, approximate age, basic personality, and speaking speed—to support consistent behavior across narrative contexts. Dialogue is generated contextually for (1) background with a single character and (2) background with two interacting characters. Spoken lines are synthesized using neural text-to-speech and presented with English subtitles. A complete list of character profiles is provided in the supplementary material.

2.5 Installation and concealment of technology

To foreground books as the primary interface, JeonStack conceals computational components within a stacked bookshelf structure. A laptop (middle tier) and projector (bottom tier) are hidden from frontal view, while a webcam is concealed inside a desk-lamp shade to capture stacking actions from above. This configuration minimizes the visibility of technical apparatus and presents the installation as a quiet reading environment in which books appear to respond autonomously.

3 Conclusion and Future Work

JeonStack demonstrates that interactive storytelling can emerge without rewriting or physically altering books. By treating existing folktales as modular narrative units and introducing stacking as a narrative grammar, the project shifts storytelling from linear reading to embodied composition. The separation between selection and stacking positions the audience not as readers of a fixed text, but as composers who assemble narrative structure through physical action.

While the current implementation allows books to be physically removed during interaction—causing the system to update based on the topmost remaining book—these changes are treated as immediate state updates rather than intentional narrative events. Future work could explicitly design book removal and reordering as meaningful interactions, enabling controlled interruption, contradiction, or narrative decay through well-defined transition rules rather than implicit sensing behavior.

References

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